

Bachelor Thesis

Identity configurations on specific classes of graph

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Bachelor student at Vietnamese - German University from 2014.

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Background:

- Algebra (VGU - 2015).
- Calculus (VGU - 2015).
- Discrete Mathematics (VGU - 2016).
- Statistics (Fra-UAS - 2016).

Outlines

Introduction

- Chip firing game

- Sandpile group

Case study: identity configuration computing

- Experimental results on specific classes of graph

- General remarks

Conclusion

Chip Firing Game

Definition [Biggs, 1999]:

- A chip-firing game on a graph G starts with a pile of tokens (chips) at each vertex.
- A configuration of the game is an assignment of chips to each vertex.
- At each step of the game a vertex v is fired, that is, chips move from v to the adjacent vertices, one chip going along each edge incident with v .
- A vertex v can be fired if and only if the number of chips currently held at v is at least $\deg(v)$, the degree of v .

Remarks: In this thesis,

- we only care about simple connected graph.
- we make sure the game is finite.

Example

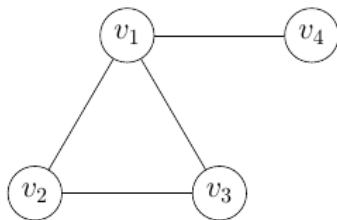


Figure: A simple connected graph G

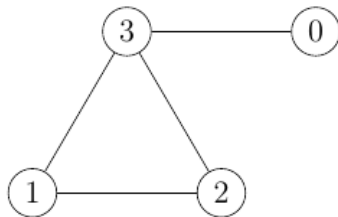
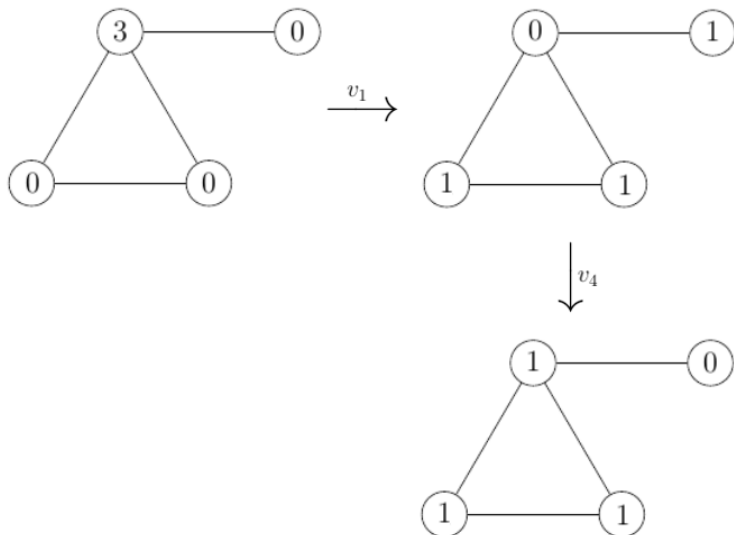
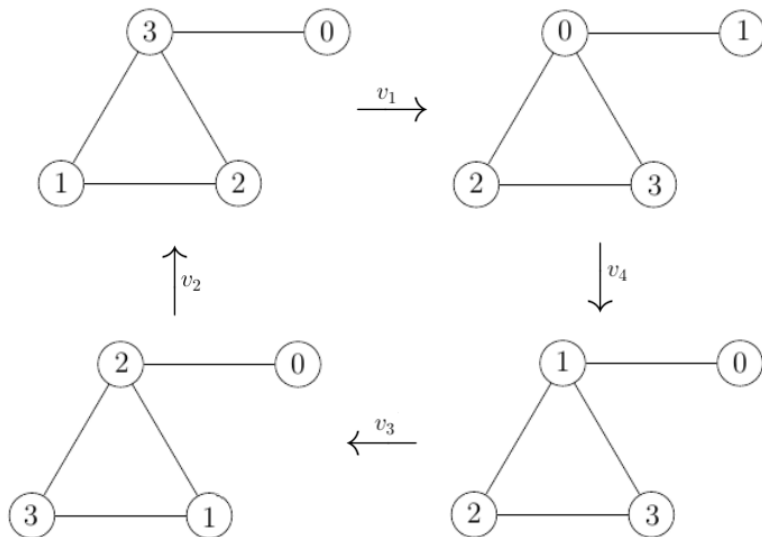


Figure: A configuration on G

Example of a finite game

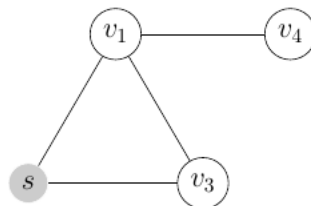


Example of an infinite game



Sink of a graph

Definition: In a simple connected graph, a vertex s is defined as a sink if it never gives out any chips to its neighbors.

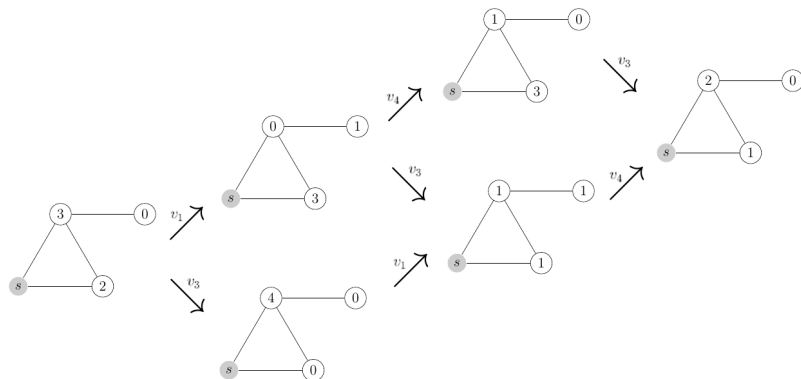


By assign a vertex to be a sink, the game is assured to be finite [Björner & Lovász, 1992, Holroyd et. al., 2008].

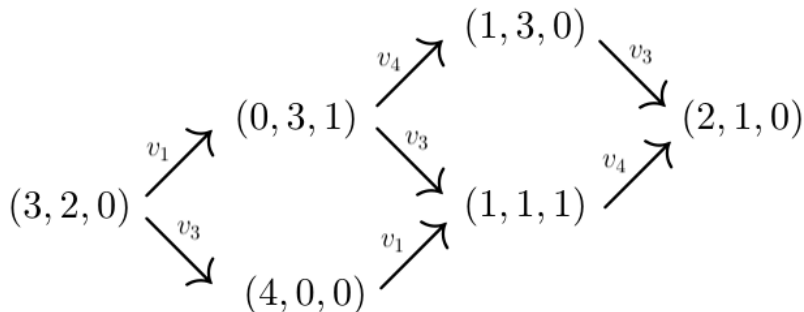
The configuration will exclude the number of chips in the sink:

$$C : V \setminus s \longrightarrow N$$

Examples of the CFG with a sink



Examples of the CFG with a sink (2)



Some new representations of a graph

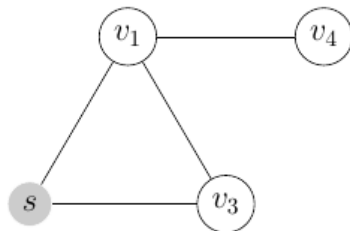
Reduced laplacian matrix:

$$\Delta = \begin{bmatrix} 3 & -1 & -1 & -1 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 2 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix} \longrightarrow \Delta' = \begin{bmatrix} 3 & -1 & -1 \\ -1 & 2 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Some interested configurations

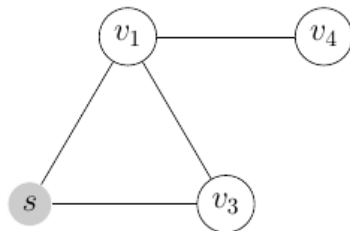
- **Stable configuration** contains no firable vertex.
- **Accessible configuration** is the one from which any other configurations can reach, given a selective adding and then selective firing chips mechanism.
- **Recurrent configuration** is stable and accessible.

Example



- Stable configurations:
 $(2, 1, 0), (2, 0, 0), (1, 1, 0),$
 $(1, 0, 0), (0, 1, 0), (0, 0, 0)$

Example (2)



- Recurrent configurations:
 $(2, 1, 0), (2, 0, 0), (1, 1, 0),$
 $(1, 0, 0), (0, 1, 0), (0, 0, 0)$

Sandpile group

The set of recurrent configurations together with the operation defined by the piece-wise addition followed by a stabilization forms an abelian group, called sandpile group of G , denoted by $S(G)$.

$$S(G) = \{(2, 1, 0), (2, 0, 0), (1, 1, 0)\}$$

A configuration I of the sand-pile group $S(G)$ is the identity if and only if for any recurrent configuration $u \in S(G)$ we have:

$$(I + u) \xrightarrow{*} u$$

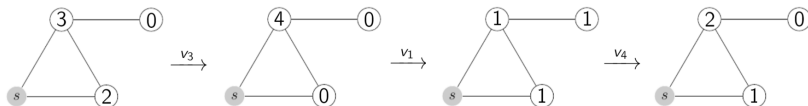
Example

Example of $(1, 1, 0) + (2, 1, 0) \xrightarrow{*} (2, 1, 0)$:

Adding:

$$c = s_1 + l = (2, 1, 0) + (1, 1, 0) = (3, 2, 0)$$

Stabilizing:



Flower graph

A flower graph of n , denoted by F_n , is a simple graph which has one vertex at the center and the others will distinctively be grouped into group of two and they, together with the common center vertex, create a 3-cycle sub-graph. Each of the 3-cycle graph is called a petal of F_n .

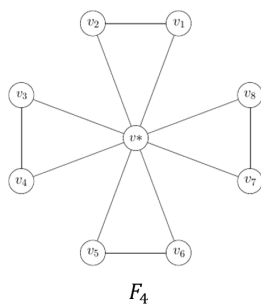
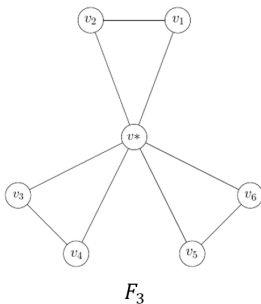
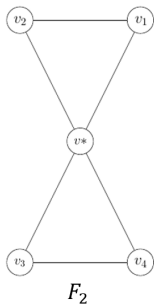


Figure: Flower graph of 2, 3 and 4 petals

Flower graph

n	F_n	$I(S(F_n))$
1	F_1	(1,1)
2	F_2	(1,1,1,1)
3	F_3	(1,1,1,1,1,1)
5	F_5	(1,1,1,1,1,1,1,1,1,1)
8	F_8	(1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1)

Table: Identity configuration on flower graphs

Remarks: The identity of flower graph is all-1.

Wheel graph

n	W_n	$I(S(W_n))$
4	W_4	(2,2,2)
5	W_5	(2,2,2,2)
6	W_6	(2,2,2,2,2)
8	W_8	(2,2,2,2,2,2,2)
10	W_{10}	(2,2,2,2,2,2,2,2,2)

Table: Identity configuration on wheel graphs

Remarks: The identity of wheel graph is all-2.

Cycle graph

k	C_{2k}	$I(S(C_{2k}))$	C_{2k+1}	$I(S(C_{2k+1}))$
1	NG	NG	C_3	(1,1)
2	C_4	(1,0,1)	C_5	(1,1,1,1)
3	C_6	(1,1,0,1,1)	C_7	(1,1,1,1,1,1)
5	C_{10}	(1,1,1,1,0,1,1,1,1)	C_{11}	(1,1,1,1,1,1,1,1,1,1)

Table: Identity configuration on cycles

Remarks: The identity of odd cycle graph is all-1.

Complete graph

n	K_n	$I(S(K_n))$
4	K_4	(2,2,2)
5	K_5	(3,3,3,3)
6	K_6	(4,4,4,4,4)
8	K_8	(6,6,6,6,6,6,6,6)
10	K_{10}	(8,8,8,8,8,8,8,8,8,8)

Table: Identity configuration on complete graphs

Remarks: The identity of odd cycle graph is all-(n-2).

Maximal stable configuration

Recall that a **maximal stable configuration** is the configuration where every vertex holds the number of chips equals to its degree minus one. From the experiments, the following observation is yielded:

Proposition

The identity configuration of flower graph, odd-cycle, wheel graph, complete graph coincides with the maximal stable configuration.

Proof.

Here is the reduced laplacian matrix of the complete graph K_n :

$$\Delta' = \begin{bmatrix} (n-1) & \dots & -1 & -1 \\ \vdots & \ddots & \vdots & \vdots \\ -1 & \dots & (n-1) & -1 \\ -1 & \dots & -1 & (n-1) \end{bmatrix}$$

Notice that the total row sum of Δ' is equal to 1. Since the maximal stable configuration of a complete graph K_n is all-(n-2), then:

$$\Delta'(K_n)(n-2) = c_{max}$$

Then from the given theorem, the complete graph accepts maximal stable configuration as its identity element of the sandpile group. □

Cycle graph

k	C_{2k}	$I(S(C_{2k}))$	C_{2k+1}	$I(S(C_{2k+1}))$
1	NG	NG	C_3	(1,1)
2	C_4	(1,0,1)	C_5	(1,1,1,1)
3	C_6	(1,1,0,1,1)	C_7	(1,1,1,1,1,1)
5	C_{10}	(1,1,1,1,0,1,1,1,1)	C_{11}	(1,1,1,1,1,1,1,1,1,1)

Table: Identity configuration on cycles

Conclusions

- The chip firing game simulation for finding the recurrent configuration.
- Some elementary results on some classes of graph.

Future works

- The sink replacement problem.
- Experiments on ear decomposition graph.

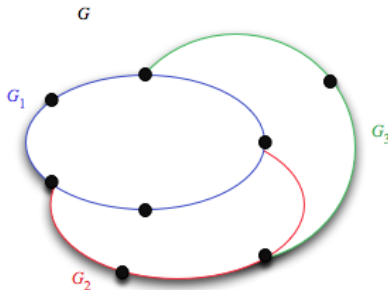


Figure: Ear decomposition graph (wikipedia.org)

References



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Björner, Anders and Lovász, László (1992)

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Dhar, Deepak (1990)

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Phys. Rev. Lett. 64, 1613 – 1616.



Holroyd, et. al. (2008)

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In and Out of Equilibrium 2. 331 – 364.

